

POSITIONS	Ross Assistant Professor The Ohio State University	Fall 2025 - present
	Adjunct Professor Cosumnes River College	Spring 2018 - Spring 2019
EDUCATION	Ph.D. in Mathematics University of California, Santa Barbara	Fall 2019 - Spring 2025
	• Thesis: <i>Skein Theory of Affine ADE Subfactor Planar Algebras</i> • Advisor: Stephen Bigelow • Certificate in College and University Teaching	
	M.A. in Mathematics University of Oklahoma	Fall 2014 - Spring 2017
	B.S. in Mathematics University of California, Davis	Fall 2010 - Spring 2014
PAPERS	5. <i>NIM-Representations of Tambara-Yamagami Generalizations</i> (with A. Czenky, E. McGovern, M. Müller, and A. Ros Camacho) Submitted. arxiv: 2602.23502 4. <i>A Mirror Deformation of Markov Numbers</i> (with L. Bittmann, P. Jouteur, E. Kantarcı Oğuz, and E. Yıldırım) Submitted. arxiv: 2602.14802 3. <i>A Well-Defined Jellyfish Algorithm for the Affine E_7 Subfactor Planar Algebra</i> Submitted. arxiv: 2601.09003 2. <i>Skein Theory of Affine A Subfactor Planar Algebras</i> Submitted. arxiv: 2410.05519 1. <i>A Determinant Formula of the Jones Polynomial for a Family of Braids</i> (with D. Asaner, S. Kumar, A. Pease, and A. Poudel) Topology and its Applications, 384 (2026).	
RESEARCH INTERESTS	Quantum algebra, low-dimensional topology, operator algebras, tensor categories, diagrammatic algebras, subfactors, classical and virtual knot theory, quantum groups, higher categories, quantum topology, representation theory, q -deformations of real numbers, cluster algebras	
AWARDS	USTARS Distinguished Graduate Student in Topology	2024
	Decolvenaere Research Accelerator Award, UCSB Graduate Division	2023
	Individualized Professional Skills (IPS) Grant, UC Santa Barbara	2023
	Math Department Graduate Fellowship, University of Oklahoma	2015
	Richard V. Andree Memorial Scholarship, University of Oklahoma	2015
	Yueh-Jing Lin Fund, UC Davis	2013
MENTORSHIP	The Ohio Summer Undergraduate Institute in Mathematics Mentor	June 2026 - August 2026
	UCSB Directed Reading Program Mentor	December 2019 - June 2023
	REU Graduate Mentor, UCSB	June 2022 - August 2022
	Women in STEM Mentor, UCSB	September 2019 - June 2022
INVITED RESEARCH STAYS	Institute of Mathematics of the Polish Academy of Sciences, The Simons Semester on "Quantum Groups and Operator Algebras"	May 2027 - June 2027
TALKS *=INVITED	*37. Moab Topology Conference; Utah State University, Moab <i>A Well-Defined Jellyfish Algorithm for the Affine E_7 Subfactor Planar Algebra</i>	May 2026

*36. UNCW Algebra and Number Theory Seminar; University of North Carolina Wilmington <i>NIM-Representations of Tambara-Yamagami Extensions</i>	February 2026
*35. Banff International Research Station Workshop; New Developments in Tensor Categories (Virtual) <i>A Well-Defined Jellyfish Algorithm for Affine E Subfactor Planar Algebras</i>	January 2026
*34. Joint Mathematics Meetings; Recent Progress in Quantum Topology Special Ses- sion; Walter E. Washington Convention Center <i>A Well-Defined Jellyfish Algorithm for Affine E Planar Algebras</i>	January 2026
*33. Joint Mathematics Meetings; Recent Developments by Women in Mathematical Physics Special Session; Walter E. Washington Convention Center <i>Fusion Categories Arising from Affine A Planar Algebras</i>	January 2026
*32. Banff International Research Station Workshop; Operator Algebraic Quantum Groups <i>A Well-Defined Jellyfish Algorithm for Affine E Subfactor Planar Algebras</i>	December 2025
*31. West Coast Operator Algebra Seminar; Pomona College <i>A Well-Defined Jellyfish Algorithm for Affine E Subfactor Planar Algebras</i>	December 2025
*30. European Quantum Algebra Lectures; Cardiff University and University of Not- tingham (Virtual) <i>Skein Theory of Affine ADE Subfactor Planar Algebras</i>	November 2025
*29. East Coast Operator Algebras Seminar; University of Nebraska - Lincoln <i>Subfactor Planar Algebras at Index 4</i>	October 2025
28. Young Mathematicians in C* Algebras; University of Southern Denmark <i>Subfactor Planar Algebras of Type Affine E7</i>	July 2025
*27. AWM Research Symposium; Non-commutative algebras, tensor categories, and diagrams Special Session; University of Wisconsin-Madison <i>Diagrammatics of Index 4 Subfactor Planar Algebras</i>	May 2025
*26. AMS Spring Western Sectional Meeting; Knot Theory and Quantum Topology Special Session; California Polytechnic State University <i>Diagrammatics of Index 4 Subfactor Planar Algebras</i>	May 2025
25. Banff International Research Station; Women in Noncommutative Algebra and Representation Theory 4 <i>Planar Algebras and their Corresponding Categories</i>	March 2025
24. Quantum Groups, Tensor Categories and Quantum Field Theory Workshop; Uni- versity of Oslo <i>Index 4 Subfactor Planar Algebras and Their Corresponding Categories</i>	January 2025
*23. Joint Mathematics Meetings; Knots, 3-manifolds, and Their Invariants Special Session; Seattle Convention Center <i>A determinant formula of the Jones Polynomial for a family of braids</i>	January 2025
22. Joint Mathematics Meetings; Advances in Operator Algebras Special Session; Seat- tle Convention Center <i>Using Diagrams to Describe Quantum Symmetries and Subfactors</i>	January 2025
*21. AMS Western Sectional Meeting; Graphical Calculus in Representation Theory and Low-Dimensional Topology Special Session; University of California, Riverside <i>Index 4 Subfactor Planar Algebras and Their Corresponding Categories</i>	October 2024
20. Quantum Groups and Representation Theory; North Carolina State University <i>Index 4 Subfactor Planar Algebras and their Corresponding Categories</i>	October 2024
19. Young Mathematicians in C Algebras; University of Glasgow <i>Using Diagrams to Describe Quantum Symmetries of Subfactors</i>	August 2024

- *18. Oberwolfach Seminar; Classification of C^* -Algebras and Dynamics Seminar; May 2024
Oberwolfach Research Institute for Mathematics
Nuclear Dimension of Cuntz-Pimsner Algebras of C^ -Correspondences*
17. Banff International Research Station Workshop; Skew Braces, Braids, and the May 2024
Yang-Baxter Equation; Banff International Research Station
Skein Theory of Subfactor Planar Algebras
- *16. USTARS Distinguished Graduate Student in Topology Speaker; April 2024
Underrepresented Students in Topology and Algebra Research Symposium; University of Iowa
Using Diagrams to Describe Quantum Symmetries of Subfactors
15. Graduate Student Topology and Geometry Conference; Michigan State April 2024
University
Skein Theory for Index 4 Subfactor Planar Algebras
14. Southeastern Lie Theory Workshop; Quantum Structures in Lie Theory; March 2024
University of Virginia
Skein Theory for Index 4 Subfactor Planar Algebras
13. Women in Mathematics in Southern California Symposium; Pomona February 2024
College
Using Diagrams to Describe Quantum Symmetries of Subfactors
- *12. Joint Mathematics Meetings; Knots, Skein Modules, and Categorification January 2024
Special Session; Moscone Center
Skein Theory for Index 4 Subfactor Planar Algebras
- *11. Claremont Topology Seminar; Pomona College November 2023
Skein Theory for Affine ADE Subfactor Planar Algebras
- *10. Twinned Conference on C^* -Algebras and Tensor Categories; Fields Institute; and November 2023
streamed live at International Centre for Mathematical Sciences
Skein Theory for Affine ADE Subfactor Planar Algebras
9. Young Mathematicians in C^* -Algebras; KU Leuven August 2023
Skein Theory of Subfactor Planar Algebras
8. Great Plains Operator Theory Symposium; The Ohio State University May 2023
Skein Theory for Index 4 Subfactor Planar Algebras
7. AMS Spring Western Sectional; Algebraic Structures in Knot Theory Special Ses- May 2023
sion; Fresno State University
Diagrammatic Presentations of Affine Type A Subfactor Planar Algebras
6. Moab Topology Conference; Utah State University, Moab May 2023
Skein Theory for Affine ADE Subfactor Planar Algebras
5. Redbud Topology Conference; University of Oklahoma March 2023
Skein Theory of Affine Type A Subfactor Planar Algebras
4. Underrepresented Students in Topology and Algebra Symposium; March 2023
University of Washington
Diagrammatic Presentations of Index 4 Planar Algebras
- *3. Quantum Groups Seminar; Université de Lille & University of Oslo (Virtual) March 2023
Skein Theory for Affine ADE Subfactor Planar Algebras
2. Tech Topology Conference; Georgia Institute of Technology December 2022
Diagrammatic Presentations of Index 4 Subfactor Planar Algebras
1. Topology Students Workshop; Georgia Institute of Technology July 2022
Diagrammatic Presentations of Affine Type A Subfactor Planar Algebras

TEACHING *=ONLINE COURSE	Instructor of Record - The Ohio State University	Fall 2025 - present
	*Math 1116 - Excursions in Mathematics	
	Math 1131 - Calculus for Business	
	Lead TA - UC Santa Barbara	Fall 2022 - Spring 2023
	Provided professional development and training to new math department TAs	
	Teaching Assistant - UC Santa Barbara	Fall 2019 - Spring 2025
	MATH 111ABC - Abstract Algebra	
	MATH 108AB - Introduction to Linear Algebra	
	MATH 8 - Transition to Higher Mathematics	
	*MATH 8 - Transition to Higher Mathematics	
	*MATH 6AB - Vector Calculus	
	MATH 34B - Calculus for Social and Life Sciences	
	Associate Instructor - UC Santa Barbara	Summer 2020, 2021, 2024
	MATH 4A - Linear Algebra with Applications	
	*MATH 6A - Vector Calculus I	
	*MATH 4A - Linear Algebra with Applications	
	Adjunct Mathematics Professor - Cosumnes River College	Spring 2018 - Spring 2019
	MATH 30 - Pre-Algebra	
	STAT 300 - Introduction to Statistics and Probability	
	MATH 20 - Arithmetic	
	Teaching Assistant - University of Oklahoma	Fall 2015 - Spring 2017
	MATH 1743 - Business Calculus I	
	MATH 1914 - Calculus 1 (Fast Track)	
	MATH 2924 - Calculus 2 (Fast Track)	
	MATH 1823 - Calculus 1	
	Instructor of Record - University of Oklahoma	Fall 2014 - Summer 2015 & 2016
	MATH 1823 - Calculus 1	
	MATH 1503 - College Algebra	
SERVICE	Operator Algebra Mentor Network Mentee Board Member	September 2025 - present
	AMS 2025 Fall Eastern Virtual Sectional Special Session Co-Organizer	February 2025 - October 2025
	Topology Seminar Co-Organizer	October 2024 - June 2025
	AWM Student Chapter at UCSB, Secretary	June 2023 - June 2025
	AWM EvenQuads 3 Data Collector	June 2024
	Directed Reading Program at UCSB Co-Organizer	December 2020 - July 2023
	Graduate Quantum Algebra and Topology Seminar Co-Organizer, UCSB	July 2023 - January 2024
	AWM Chapter at UCSB Founder and President	August 2022 - June 2023
	Quantum Algebra and Topology Seminar Co-Organizer, UCSB	September 2021 - June 2022
	AWM Essay Contest Judge	January 2020 - March 2020
	Math Teacher's Circle - Oklahoma City Founder and Instructional Leader	April 2015 - June 2017
WORKSHOPS	Institute of Science and Technology Austria (ISTA)	
	Women in Topology V	January 2027
	International Centre for Mathematical Sciences (ICMS)	
	Women in Mathematical Physics 3	April 2026
	Oberwolfach Research Institute for Mathematics (MFO)	
	Classification of C^* -Algebras and Dynamics Seminar	May 2024

Banff International Research Station (BIRS)

Operator Algebraic Quantum Groups
Women in Noncommutative Algebra and Representation Theory 4
Skew Braces, Braids, and the Yang-Baxter Equation Workshop
Women in Operator Algebras 3

November 2025
March 2025
May 2024
June 2023

American Institute of Mathematics (AIM)

Quantum Invariants and Low-dimensional Topology

August 2023

**TRAVEL
GRANTS**

National Science Foundation Travel Grant
AWM Symposium Travel Grant
GSA Conference Travel Grant, UC Santa Barbara
UCSB Doctoral Student Travel Grant
AMS Sectional Travel Grant
GSA Conference Travel Grant, UC Santa Barbara
Operator Algebra Mentor Network Grant

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